



Swinburne University of Technology Sarawak Faculty of Engineering, Computing and Science

School of Information and Communications Technologies

COS10005 Web Development Assignment 1 - Semester 2, 2026

Late submission penalty is 5% of the total available marks per day
This is an Individual assignment.

Contribution to Final Assessment: 30%

Purpose of the assignment

This is an individual assignment to familiarize students with the techniques and skills involved in creating webpages using validated HTML5 and validated CSS; loading and testing the website on the server; and demonstrating an understanding of HTML5 elements and attributes, CSS selectors and properties and their proper use. Moreover, it is to provide students with an appreciation of the importance of the separation of structured content from presentation as discussed in the lectures.

No JavaScript is to be used in this assignment.

Case Study Scenario:

Malaysia represents one of the world's notable biodiversity hotspots, with a large number of remarkable species and ecosystems, but the region is also one of the most biotically threatened. Population growth and the resulting need for agricultural production contribute to increasing pressure on natural resources and ecosystems. These new and greater pressures, exerted by the results of human activities, require us to rethink the way we manage and conserve biodiversity. In recent years, the emergence of initiatives based on **citizen science** approaches to biodiversity monitoring has profoundly transformed the way we oversee biodiversity.

Citizen science, an alliance between professional scientists and the general public in scientific research, holds immense potential to augment both the "amount of data" and the "number of contributors" in tropical regions. As part of this project, an alpha version of the citizen science platform is proposed, empowering users with **web interfaces** to identify Endangered, Rare and Threatened (ERT) plants through an **automatic visual identification engine**. In addition, the web application will integrate a function akin to a plant encyclopedia, enabling users to explore encountered plant species.

In this project, you are required to come out with the web pages for user/visitor and admin (in assignment 2).

More information can be found in the case study document.

To highlight this, the assignment is specified and evaluated in four sections.

- The development of validated HTML5 webpages.
- The development of validated CSS files to present webpages on various devices.
- The development of accessible webpage.
- The implementation of CSS animations.

Section One: HTML5 (40 marks)

Content and presentation of Web Pages

1. Introductory homepage (index.html)

This page should contain appropriate title and graphic related to the project. This page should also contain a menu that links to the other pages on your website. This menu should appear on every page of your website.

2. Explorer page and pages (explore.html, explore1.html, explore2, html, explore3.html, etc.)

Minimum 4 pages, create a page to link to the other 3 pages (families, genera, and species). The 3 pages to be included on the website to describe the following 3 categories:

- Families
- Genera
- Species

Each page describes a specific sub-category. The following requirements specify the plants to be included in each sub-category (please refer to PlantsData document):

- Families (2 Families)
- Genera (2 Genera for each Family)
- Species (2 Species for each Genus)

At least one appropriate image must be displayed for each plant. The content should be appropriately structured with headings, sections, asides, lists etc.

Each page should be sufficiently described in English (at least 200 words). Use your own words. Do not copy and paste from other websites.

The pages must contain:

- Hierarchically structured headings of at least 2 levels (e.g. make sure your H2 is below an H1).
- More than one <section>.
- An <aside> with appropriate content, i.e. a section of a page that consists of content that is tangentially related to the content.
- At least one appropriate related to your topic. This image should be in a <figure> element with <caption>.
- At least one ordered list.
- At least one definition list.

Please refer to “Sources/References” below.

3. Identify page (identify.html) and

4. Contributor page (contributor.html)

The content should be appropriately structured with headings, sections, asides, lists etc.

The pages must contain:

- Hierarchically structured headings of at least 2 levels (e.g. make sure your H2 is below an H1).
- More than one <section>.
- An <aside> with appropriate content, i.e. a section of a page that consists of content that is

tangentially related to the content.

- At least one appropriate related to your topic. This image should be in a <figure> element with <caption>.
- At least one ordered list.
- At least one definition list.

Please refer to “Sources/References” below.

5. Contribution form (contribute.html)

This page has a form that allows user to contribute plant data to the website. The form will have a number of input fields. The name, type and data format of these input fields are as follows (you are allowed to modify to match your business type).

- a. **Name of the Contributor:** *text* type, maximum of 25 characters, alphabetical characters only
- b. **Plant's name:** *text* type, maximum of 25 characters, alphabetical characters only
- c. **Photo(s):** *file* type
- d. **Comments:** *text area* type, setting the rows and columns according to the form design.

Please refer to the “Forms values required” below.

6. Enquiry page (enquiry.html)

This page has a form that allows user to make an enquiry. The form will have a number of input fields. The name, type and data format of these input fields are as follows (you are allowed to modify to match your business type).

- a. **First name:** *text* type, maximum of 25 characters, alphabetical characters only
- b. **Last name:** *text* type, maximum of 25 characters, alphabetical characters only
- c. **Email address:** *email* type
- d. **Phone number:** *text* or *tel* type, maximum of 10 digit. Use a placeholder.
- e. **Enquiry:** use a *select* type to select from the list (products/services).
- f. **Comments:** *text area* type, setting the rows and columns according to the form design.

Please refer to the form submission format below.

Sources / References:

- You can refer to materials (not copy and paste) and images from other web sites. In this case, the source of all materials must be acknowledged. This acknowledgement should be immediately after the material and include a hyperlinked URL to the original source. The text of the hyperlink reference can be a short name but the hyperlink must work.

Form submission format.

The user should not be able to submit the form if any of the required fields is blank. Use HTML5 form validation to check the format and completeness of the form.

Forms values required

All form values ‘required’. **The user should not be able to submit the form if any of the required fields is blank.** Use HTML5 form validation to check the format and completeness of the form.

7. Profile Individual page

One profile Web page must be created. Information in these Web pages **must be consistently formatted** with CSS in the following way:

Information	Format requirement
Your name	Centre aligned to the page, bold, font size 200%
Student number	Arial font (default to San Serif if Arial not available)
Course you are doing	Colour specified as a RGB number
Photo of you	- Floating - Groove border with 20px padding - Size in bytes less than 100k
A mailto link to their email.	Centred at the bottom of page in the footer

It should also include personal profile, such as resume, interests, or information that is related to you. This extra information gives you an opportunity to extend the techniques you apply in your assignment, and should include:

- Demographic information about you
- Description of hometown
- A great achievement in your life (so far)
- A list of your favorite books, music, movies, or games

Using a table for the extra information given above with the following required formatting:

- Centred Headings in bold
- Table cells have a background colour specified in hexadecimal number

The following HTML elements **MUST** be used in this assignment. Some will be used on all pages while others might only be used once.

- General elements - <head>, <title>, <meta>, <body>
- Structural elements - <header>, <nav>, <main>, <footer>, <aside>, <article>, <section>, <aside>
- Content elements - <h#>, <p>, <hr/>
 - , , <table>, and <a>
- Form elements - <form>
 - <input>
 - <label>

Other HTML elements such as <div>, , <blockquote>, , , <fieldset> and more can be used as necessary and appropriate for the content.

A menu must be created on each web page that allows users to navigate through all the webpages on your website. This menu must be properly styled using CSS.

Please organize your HTML code properly and annotate the source code with comments as necessary.

Section Two: Basic CSS (20%)

Develop one CSS files that style the web pages, one for a desktop and another for a mobile phone. The webpages must be styled to suit the corresponding device, e.g., the layout, font size and the sizes of the major components of the webpages.

Demonstrate a range of selectors, including but not limited to:

- Element *e.g., h1{}, p{}*
- Class *e.g., .image{}, highlighted{}*
- Id *e.g., #tbName{}*
- Grouping *e.g., h1, h2, h3{}*
- Contextual *e.g., ul li{}*
- Pseudo-classes *e.g., :hover{}*

Demonstrate the following groups of CSS properties:

- Menu *e.g., style a navigator that contains links to all web pages*
- Web Typography *e.g., fonts, colors, text alignment*
- Box Model and Visual Formatting *e.g., border, margin, padding, width, height, background image*
- Liquid Page Layout *i.e., page layout adapts as you resize the browser window*

Use of other CSS selectors and CSS properties are encouraged, as necessary and appropriate for styling the webpages.

Please properly organized your CSS code and include comments within your CSS code as needed. E.g., `/* banner style */`.

Creativity is strongly encouraged in the web site design. Creative designs will be rewarded with bonus marks.

Section Three: Enhancement (20%)

You can develop enhancements on your Web site that go beyond the requirements specified above. This is an opportunity to demonstrate your ability to discover techniques from a range of sources and apply them in a standards compliant manner.

These enhancements need to be implemented on your *existing* Web pages (index.html, <product pages>.html, order.html, and <about me pages>.html). The extra feature needs to enhance your Web site in a relevant way.

On a separate Web page called **enhancement1.html** list and describe each enhancement you have made and how you have significantly extended the basic HTML and CSS beyond what is covered in the Labs. Hyperlink from the list to where the feature is implemented on your Web site. If it is a CSS feature, hyperlink to an example of the html that is selected by the CSS rule.

Examples of HTML/CSS enhancements you might make that will contribute a higher mark include:

- Embedding External Content in your webpage i.e. Google Maps (about.html), related Youtube video, Photo gallery engine, widgets etc.
- Effective and appropriate use of other HTML5 elements not covered in tutorials (e.g. Canvas, Video, Audio). Note: some HTML5 elements may not be fully supported by Firefox, so make sure you check first as the assignment will be tested on Firefox.
- An additional CSS file(s) that presents your Web site specifically for mobile phone / tablet sized displays.

- Additional CSS properties or selectors (e.g. support for interactivity) not covered in the tutorials. For example the use of CSS pseudo-elements and classes, child or siblings combinators, attribute selectors, etc.
- Additional comments in the CSS files to show how to edit the declaration values to, for example, change the colour scheme. Comments must be in the context of its application, and not general comments.
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To gain a full 20 marks for a feature it must:

- Be relevant to / enhance the content of the Web site
- Be well described (as explained above)
- Be non-trivial. Judging this will be at the sole discretion of your tutor/marker. As a guide if the enhancement has only taken a couple of lines of code it is likely to be trivial.

A maximum of 4 enhancements will be assessed.

Section Four: Project Report (20%)

Component	Content
Title	• Report title • Your name • Student ID
Introduction	• Project introduction • Objective of the report • Outline of the report's structure
Website content	• Introduce the purpose of each page and main content of each page • Some technical details on how you use HTML markups • A site map
Website style	• Introduce the presentation of your website • Some technical details on how you use CSS markups • Screenshots to show the design/user interface
Key features	• Highlight the key/innovative features of the website • Describe your enhancement (if there is any)
Conclusion	• Summary of the report • Recommendations for future improvements
References	• (Optional) List of reference materials if used
Appendix	• (Optional) Information that supports but is not essential to the report

Web Site Folder Structure and Deployment Requirements

Create a website structured as described below. You can create additional HTML files for your content (depending on what your content requires), but the following is needed:

assign1/	<i>You must have this folder – case sensitive!</i>
index.html	
product1.html	<i>etc...</i>
order.html	
aboutme1.html	<i>etc...</i>
enhancements.html	
...other html pages	<i>(if any)</i>
images/	<i>Folder for images for your page content</i>
styles/	<i>Folder for style.css other css files</i>
styles/images/	<i>Folder for images referred to by your css files e.g. background</i>
xml/	<i>Folder for the XML version of the webpages</i>

Notes:

- HTML files should only be in the root “yourname_assign1/” folder – not anywhere else.
- All images used for the content should be stored in the “assign1/images/” folder.
- There should be a “style.css” file in the “assign1/styles/” folder.
- All links to your files (CSS or images) should be relative. Do not use absolute links, as these links will be broken when files are transferred for deployment/marketing. No marks will be allocated for broken links.

Assignment Submission

An electronic copy of your assignment should be submitted through Canvas at <https://www.swinburne.edu.my/canvas/> on or before the due date.

- Make sure all your files are in the correct folders and compress your root folder contains all your sub-folders with HTML, CSS, and images into a zip file named “**yourname_assign1.zip**”.
- You can submit a maximum of 3 times through Canvas. The latest attempt will be marked.
- Note that all deliverables must be submitted as softcopy. You are only required to submit the hardcopy of your report, marking scheme and assignment cover sheet.